

Cloud Computing Case Study – Online Learning Platform

Section A: Understanding the Need for Cloud

Q1. Explain why cloud computing is suitable for the given startup

Ans:

- **Scalability:** The platform can automatically handle massive traffic spikes during exam seasons and scale down during breaks.
- **Cost-Efficiency:** With a limited initial budget, the startup avoids heavy upfront costs for physical servers.
- **Speed to Market:** Cloud services allow the startup to deploy video streaming and analytics tools quickly without building infrastructure from scratch.

Section B: Cloud Deployment Model Analysis

Q2. Choosing the Cloud Type

a) Which cloud type would you recommend for this startup?

- Public Cloud
- Private Cloud
- Hybrid Cloud

Ans: Public Cloud

b) Justify your choice based on:

- **Cost:** No capital expenditure (CapEx); you only pay for what you use.
- **Scalability:** Offers near-limitless resources to handle fluctuating user traffic
- **Security:** High-level security is managed by the provider (e.g., AWS, Google Cloud), which is sufficient for a starting platform.
- **Operational overhead:** The provider manages the hardware and maintenance, letting the startup focus on building the app.

Q3. Business Growth Scenario

After two years, the startup:

- Handles sensitive student data
- Partners with universities
- Requires higher compliance and control

a) Would you continue with the same cloud type?

Ans: No, I would likely transition to a Hybrid Cloud.

b) If not, which cloud type would you suggest and why?

Ans: As the startup now handles sensitive data and requires higher compliance, a Hybrid Cloud allows them to keep sensitive student data on a Private Cloud (for control) while keeping the public-facing learning app on the Public Cloud (for scale).

Section C: Cloud Service Model Evaluation

Q4. Choosing the Service Model

a) Which cloud service model is most suitable for developing this platform?

- IaaS
- PaaS
- SaaS

Ans: (PaaS- Platform as a Service)

b) Explain why the other service models are less suitable for this use case.

Ans:

- IaaS: Requires too much manual management of operating systems and networking, increasing operational overhead.
- SaaS: While easy, it wouldn't allow the startup to build their own custom quizzes, analytics, and unique platform features

Q5. Shared Responsibility Model

For the selected service model, specify who is responsible (Cloud Provider or User) for managing the following:

Components	Responsible Party
• Infrastructure	Cloud Provider
• Operating System	Cloud Provider
• Application code	User(Startup)
• Scaling	Cloud Provider(Automated)
• Security patches	Cloud Provider(For OS/Infra)

Section D: Scalability & Performance

Q6. Handling Fluctuating Traffic

Explain how cloud computing helps the platform handle:

- Sudden spikes in user traffic

Ans: Handled by Auto-scaling, which automatically adds more server instances when exam-time traffic increases.

- Reduced traffic during off-peak periods

Ans: Handled by Elasticity, which removes unnecessary resources during off-peak times to save money.

Mention relevant cloud features in your answer.

Ans:

- **Auto-scaling:** The ability to automatically adjust resource levels based on real-time demand.
- **Elasticity:** The power of the infrastructure to expand and contract seamlessly as needed.
- **Load Balancing:** A feature that distributes incoming student traffic across multiple servers to ensure no single server is overwhelmed

Section E: Cost Optimization

Q7. Cost Management in Cloud

a) Explain how the pay-as-you-go pricing model benefits this startup

Ans: Pay-as-you-go Benefits:

- Eliminates the risk of paying for idle servers during low-traffic months.
- Allows the startup to reinvest saved capital back into platform development rather than hardware.